PROJECT DOCUMENTATION

Medical Diagnostic Center

ServiceNow Service Portal Application

Built on the ServiceNow Platform

Service Portal | Business Rules | GlideRecord | ACLs

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1. Introduction

1.1 Project Title

Medical Diagnostic Center – ServiceNow

1.2 Team Members

S.No.	Team Member	Role / Responsibility
1	K.Sai Dinesh	Project Lead / ServiceNow Developer

1.3 Technology Used

The project is developed using the ServiceNow Platform and ServiceNow Service Portal. The application uses custom ServiceNow tables, Service Portal widgets, JavaScript, AngularJS, Business Rules, Client Scripts, GlideRecord, ServiceNow authentication, roles, ACLs, and email notifications.

2. Project Overview

2.1 Purpose

The purpose of the Medical Diagnostic Center project is to provide a centralized digital platform for patients to search for diagnostic tests, select a test, book an appointment, provide patient details, complete the payment process, track appointment status, and access laboratory reports.

The system reduces dependency on manual processes by providing an integrated workflow through the ServiceNow platform.

2.2 Key Features

- Diagnostic Test Catalog
- Test Selection
- Patient Details Management
- Online Appointment Booking
- Appointment Date and Time Selection
- Payment Status Management
- Booking Confirmation
- My Appointments
- Appointment Status Tracking

- Appointment Cancellation
- Automatic Laboratory Report Generation
- My Lab Reports
- Report Details and Findings
- Search, Sort and Filter
- Notifications
- Role-Based Access Control
- ServiceNow Business Rule Automation

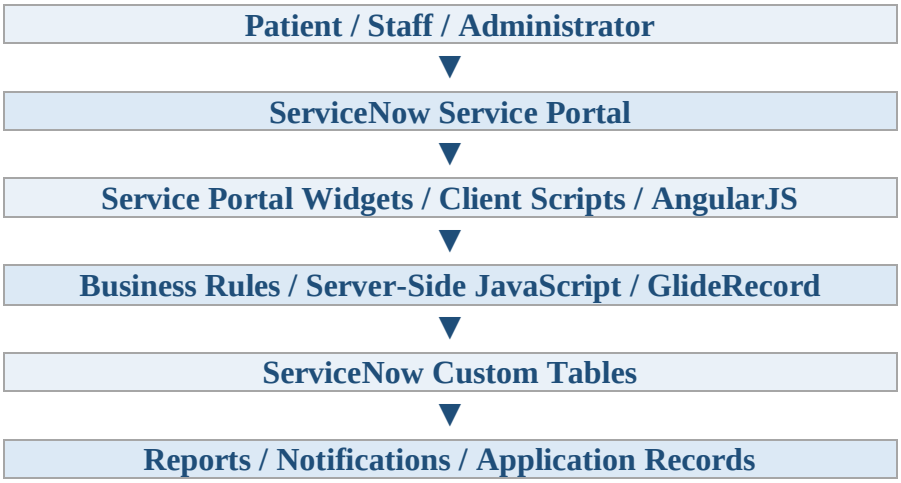
2.3 Main User Workflow

> **Browse Tests → Select Test → Enter Patient Details → Book Appointment → Payment → Confirmation → Test Completed → Laboratory Report Generated → View Report**

3. Architecture

3.1 System Architecture

The Medical Diagnostic Center follows a ServiceNow-based layered architecture.



Architecture Components

Layer	Technology / Component	Purpose
User Interface	ServiceNow Service Portal	Provides the user-facing application
Frontend	HTML, CSS, JavaScript, AngularJS	Creates interactive portal pages and widgets
Application Logic	Business Rules, Client Scripts	Implements application workflows and validations

Layer	Technology / Component	Purpose
Server Logic	JavaScript, GlideRecord, GlideSystem	Reads, creates and updates ServiceNow records
Database	ServiceNow Platform Database	Stores application data
Security	Roles, ACLs, Authentication	Controls access to application data
Notifications	ServiceNow Email Notifications	Sends booking/status/report notifications
Hosting	ServiceNow Cloud Platform	Hosts the application
Development Environment	ServiceNow PDI	Used for development and testing

3.2 Database / Tables

The application uses the following custom ServiceNow tables:

Table	Purpose
x_1894577_medica_0_patient	Stores patient information
x_1894577_medica_0_appointment_table	Stores appointment information
x_1894577_medica_0_diagnostic_test	Stores diagnostic test information
x_1894577_medica_0_report_table	Stores laboratory reports
sys_user	Stores ServiceNow user information

Table Relationships

- Patient records are associated with appointments.
- Appointments are associated with diagnostic tests.
- Completed appointments are associated with laboratory reports.
- Reports contain findings, report number, report date, status and verification information.
- ServiceNow users are used for authentication and user association.

4. Setup Instructions

4.1 Prerequisites

The following are required:

- ServiceNow account
- ServiceNow Personal Developer Instance (PDI)
- Supported web browser
- ServiceNow application development access
- Basic knowledge of ServiceNow configuration
- JavaScript knowledge for scripting and Business Rules

4.2 Application Setup

1. Log in to the ServiceNow Personal Developer Instance.
2. Open Application Studio / Studio.
3. Create or open the Medical Diagnostic Center application.
4. Create the required application tables.
5. Configure table fields and reference relationships.
6. Configure Service Portal pages.
7. Create the required Service Portal widgets.
8. Configure Client Scripts and Business Rules.
9. Configure roles and ACLs.
10. Configure email notifications.
11. Insert sample diagnostic test records.
12. Test the complete patient workflow.
13. Verify laboratory report generation.
14. Validate the application through UAT.

Note: The application runs entirely on the ServiceNow platform. No separate software installation, database setup, or external server configuration is required.

5. Application Structure

The application is developed on the ServiceNow platform. Its structure is organized into Service Portal components, which form the user interface, and application logic components, which implement the server-side behavior.

5.1 Service Portal Structure

Component	Purpose
Portal	Main Medical Diagnostic Center portal
Pages	Test Catalog, My Appointments, My Lab Reports and related pages
Widgets	Display tests, appointments, reports and interactive UI
Client Scripts	Handle client-side interaction and navigation
HTML	Widget/page structure
CSS	User interface styling
AngularJS	Dynamic portal behavior

5.2 Application Logic Structure

Component	Purpose
Business Rules	Server-side automation
GlideRecord	Database record operations
Client Scripts	Client-side validation and interaction
Email Notifications	Booking and status notifications
ACLs	Data access control
Roles	User authorization

Important Business Rule

An important automation is implemented on the Appointment table. When an appointment reaches the Completed status, the Business Rule automatically creates a corresponding laboratory report if a report does not already exist.

6. Running the Application

The Medical Diagnostic Center runs entirely on the ServiceNow platform and is accessed through the configured Service Portal. The steps below describe how to run and verify the complete workflow.

6.1 Steps

1. Start the ServiceNow PDI.
2. Log in to the ServiceNow instance.

3. Open the configured Service Portal.
4. Navigate to the Medical Diagnostic Center portal.
5. Open the Test Catalog.
6. Select a diagnostic test.
7. Continue through the appointment booking process.
8. Complete the payment step.
9. Verify the appointment in My Appointments.
10. Complete the appointment from the appropriate ServiceNow interface.
11. Verify automatic report generation.
12. Open My Lab Reports to view the generated report.

6.2 Portal Access

The project is accessed through the ServiceNow instance's configured Service Portal URL.

7. Application / API Documentation

Service Portal widgets communicate with their server-side widget scripts, which use ServiceNow APIs such as GlideRecord to retrieve and update records. The server-side operations supported by the application are listed below.

7.1 Server-Side Operations

Operation	Purpose
Retrieve Patient	Identifies the logged-in patient's record
Retrieve Appointments	Gets appointments associated with the patient
Retrieve Diagnostic Tests	Gets available tests
Create Appointment	Creates a new appointment record
Update Appointment	Updates appointment status/details
Cancel Appointment	Cancels an eligible appointment
Retrieve Reports	Gets laboratory reports
Generate Report	Business Rule creates report after completion

7.2 Example ServiceNow Server-Side Logic

```
var gr = new GlideRecord('x_1894577_medica_0_appointment_table');
gr.addQuery('created_by', gs.getUserID());
gr.query();

while (gr.next()) {
   // Process appointment record
}
```

This uses GlideRecord to interact with ServiceNow database records.

8. Authentication and Authorization

8.1 Authentication

The application uses ServiceNow authentication to identify users accessing the Service Portal. The logged-in ServiceNow user can be identified using the current user session.

8.2 Authorization

Access is controlled using:

- ServiceNow Roles
- Access Control Lists (ACLs)
- Table-level access
- Record-level access
- User permissions

This ensures that users can access only the information and functionality permitted to their role.

8.3 Security Flow

> **User Login → ServiceNow Authentication → Role/ACL Validation → Authorized Portal/Data Access**

9. User Interface

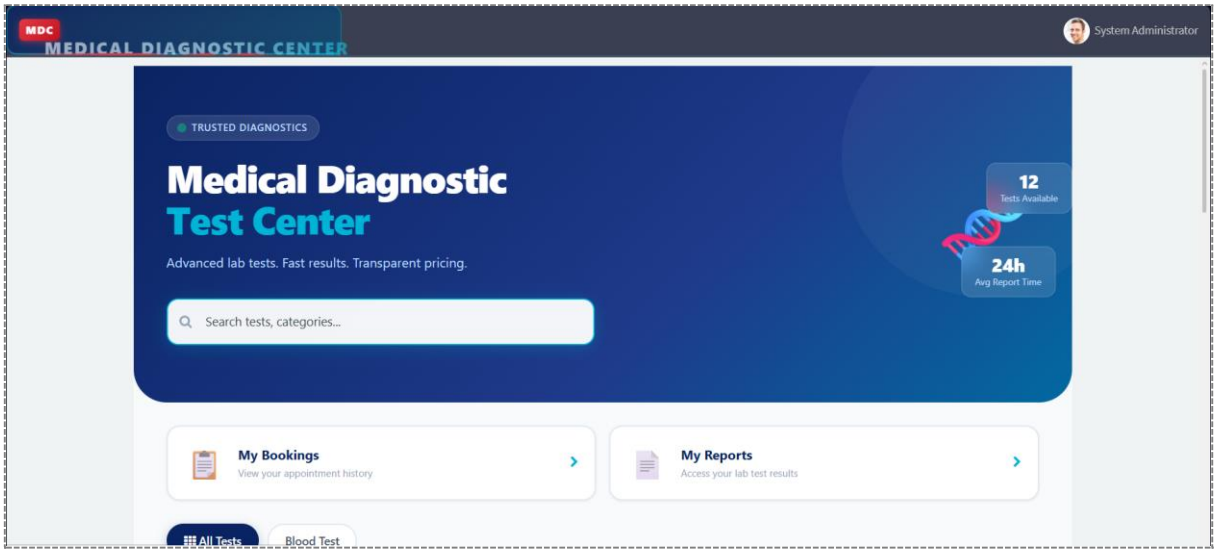
The Medical Diagnostic Center provides a ServiceNow Service Portal interface designed for patient interaction.

9.1 Major UI Screens

1. Home / Test Catalog

Displays available diagnostic tests and allows the patient to select a test.

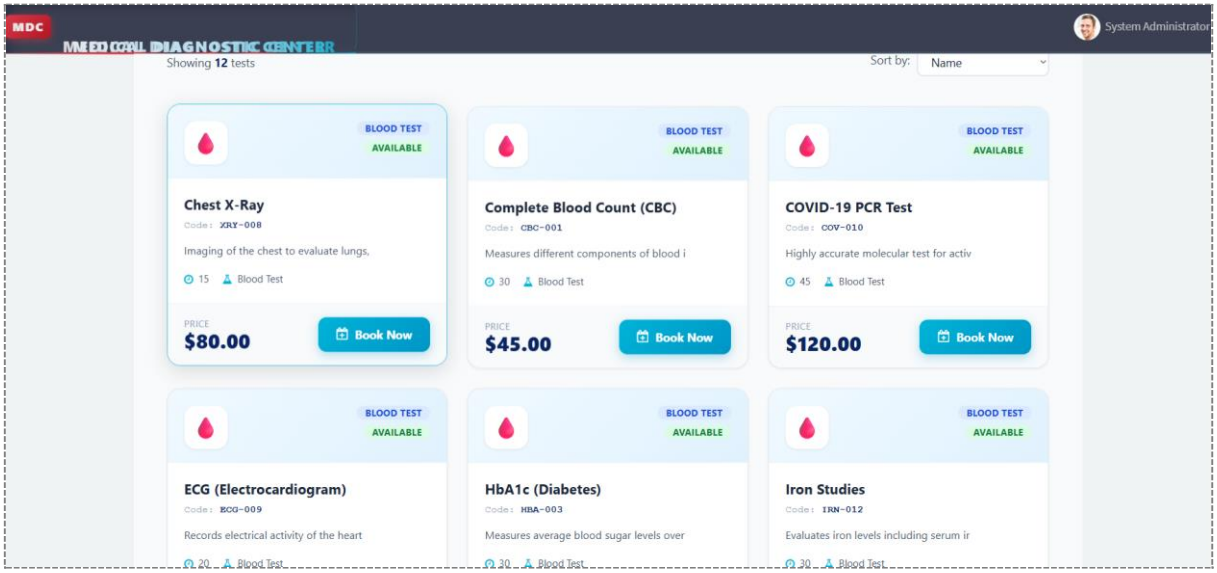
Figure 1: Medical Diagnostic Center Home / Test Catalog



2. Diagnostic Test Selection

Allows the patient to review the details of a diagnostic test and select it for booking.

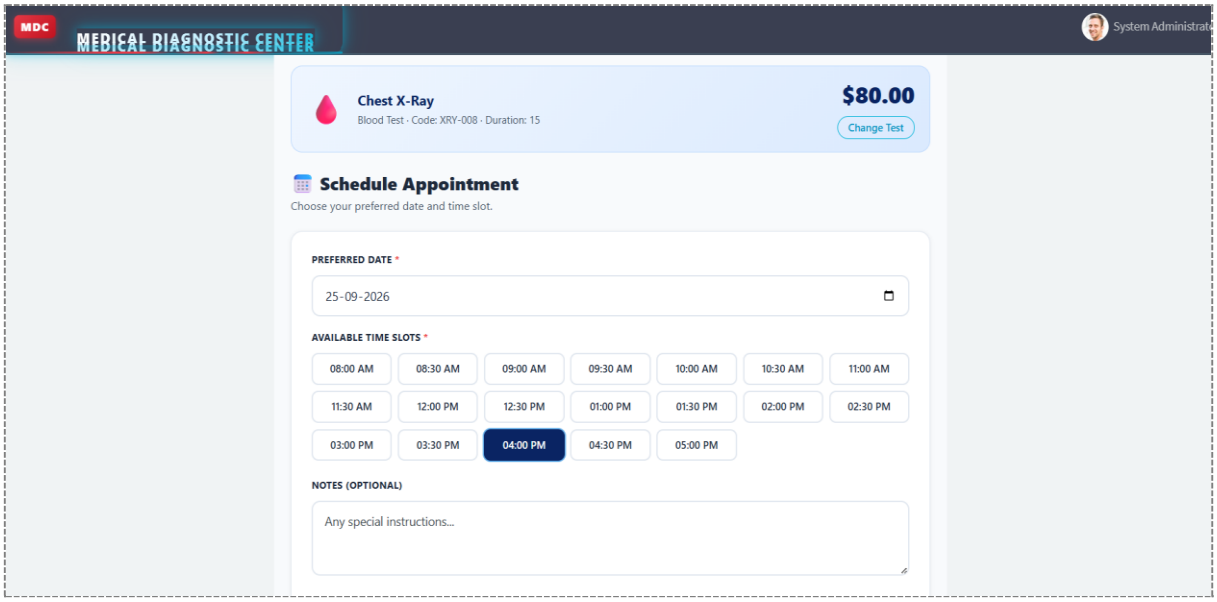
Figure 2: Diagnostic Test Selection



3. Appointment Booking

Allows the patient to enter details and select appointment information.

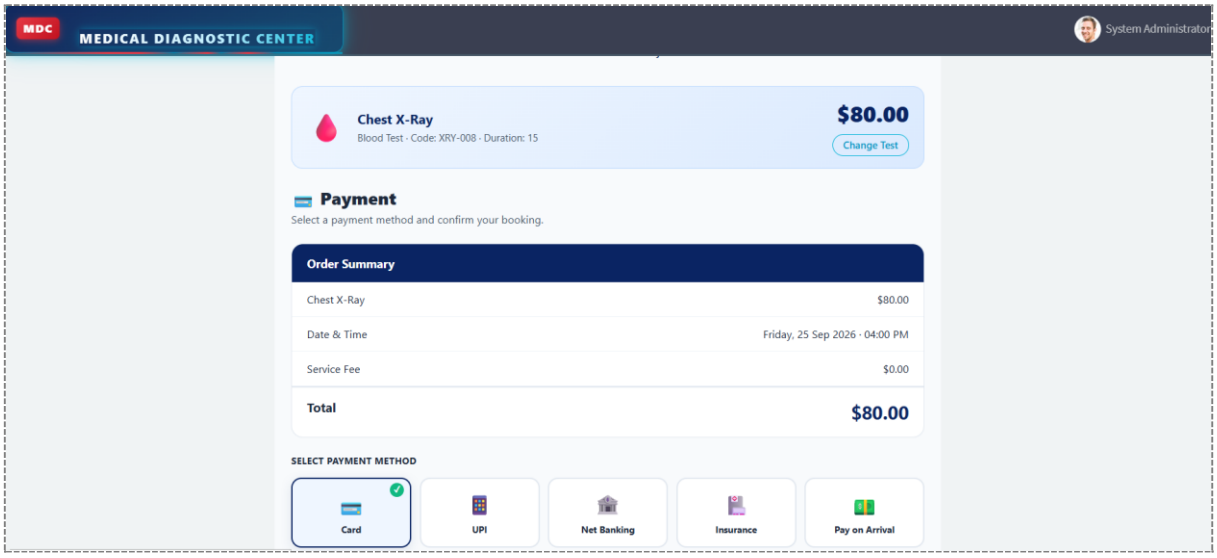
Figure 3: Appointment Booking



4. Payment

Displays the payment step and payment status.

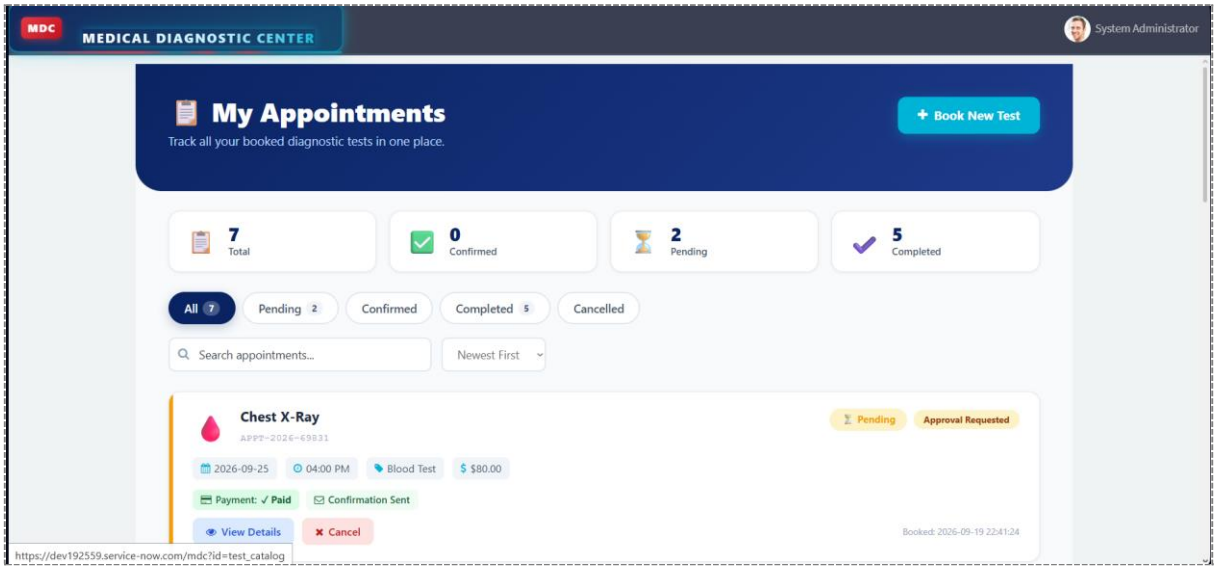
Figure 4: Payment / Confirmation



5. My Appointments

Displays the patient's appointments, status, search and sorting options.

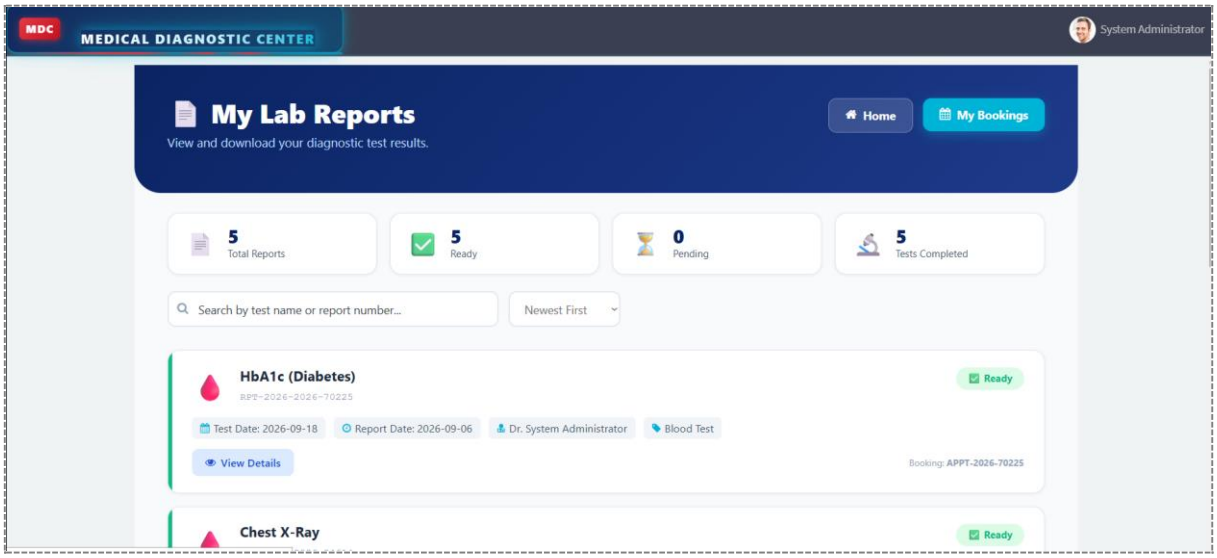
Figure 5: My Appointments



6. My Lab Reports

Displays available laboratory reports and report status.

Figure 6: My Lab Reports



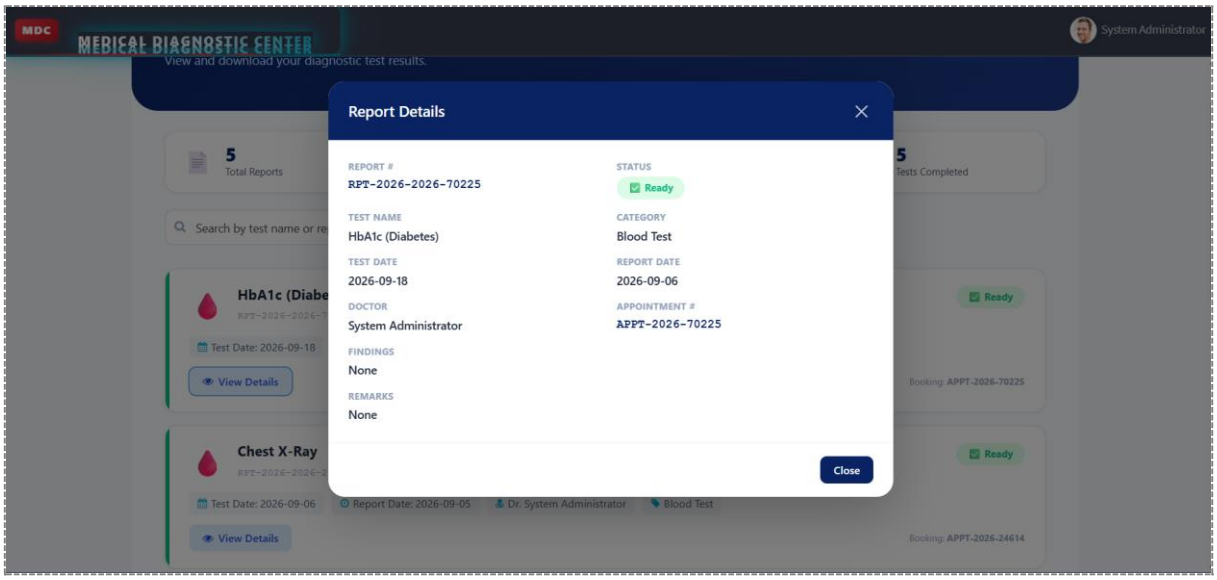
7. Report Details

Displays the following report information:

- Report Number
- Diagnostic Test
- Report Date
- Status
- Verified Doctor

- Findings

Figure 7: Report Details



10. Testing

10.1 Testing Strategy

The application was tested using functional, system, integration and User Acceptance Testing (UAT).

10.2 Functional Testing

S.No.	Test Area	Result
1	Test Catalog	**PASS**
2	Test Selection	**PASS**
3	Patient Details	**PASS**
4	Appointment Booking	**PASS**
5	Payment Status	**PASS**
6	Appointment Tracking	**PASS**
7	Appointment Cancellation	**PASS**
8	Automatic Report Generation	**PASS**
9	Laboratory Report Viewing	**PASS**
10	Search and Sorting	**PASS**

S.No.	Test Area	Result
11	Notifications	**PASS**
12	Security / Access Control	**PASS**

10.3 UAT Result

Metric	Result
Total Test Cases	14
Passed	14
Failed	0
Not Tested	0
Overall Result	PASS

11. Screenshots / Demo

This section contains the supporting screenshots for the application workflow, configuration and automation.

11.1 Demo Workflow

Login → Browse Tests → Select Test → Enter Details → Book Appointment → Payment → Confirmation → Complete Test → Generate Report → View Lab Report

11.2 Additional Screenshots

Figure 8: Patient Details

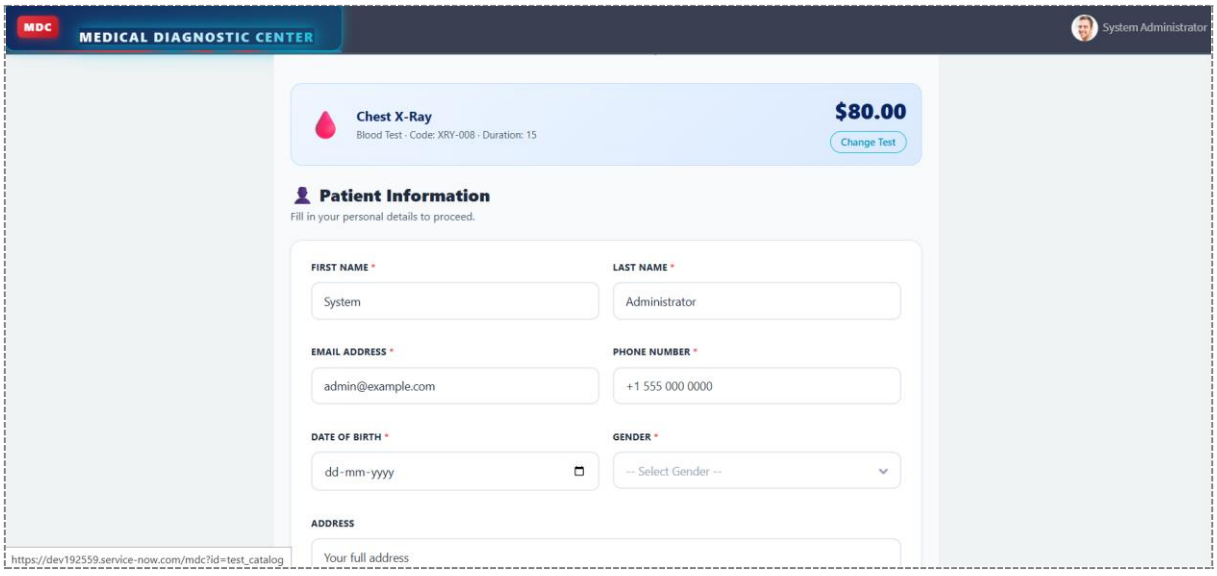


Figure 9: Payment Status

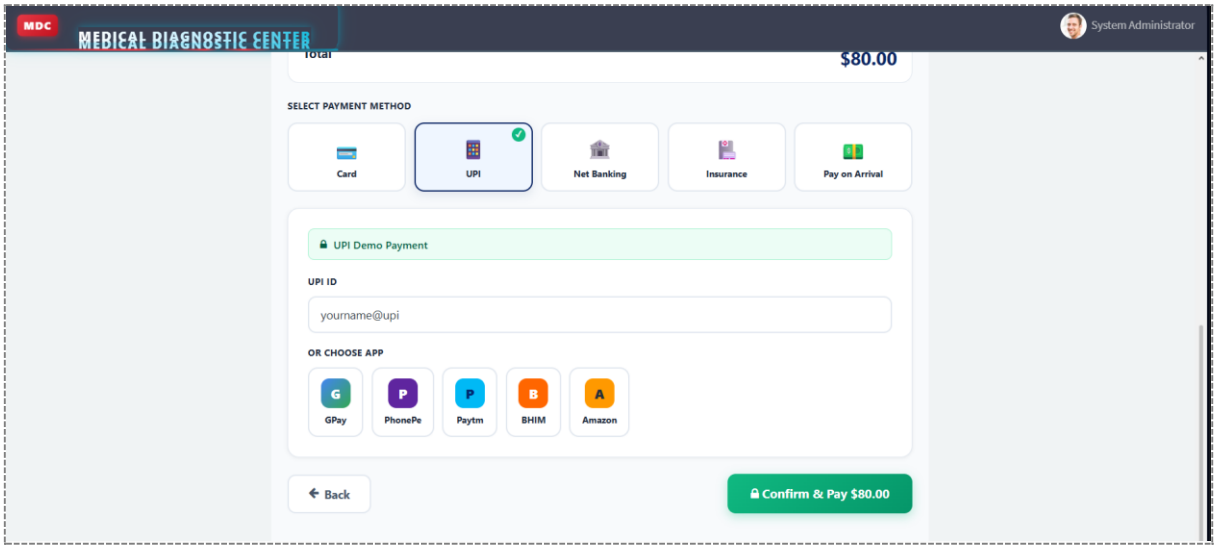


Figure 10: Booking Confirmation

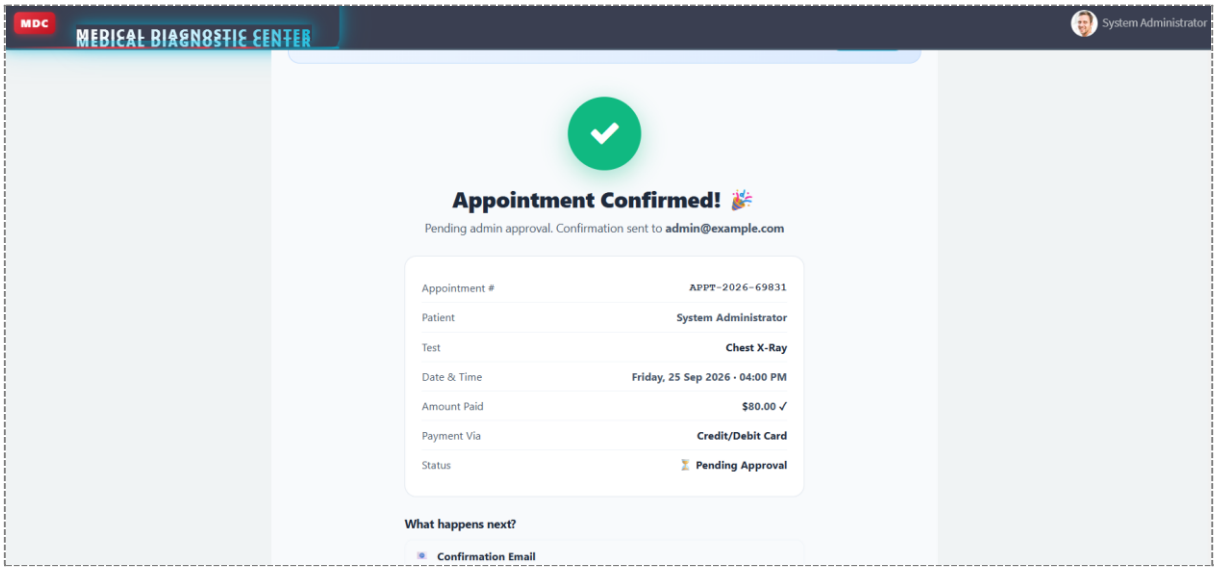


Figure 11: Appointment Cancellation

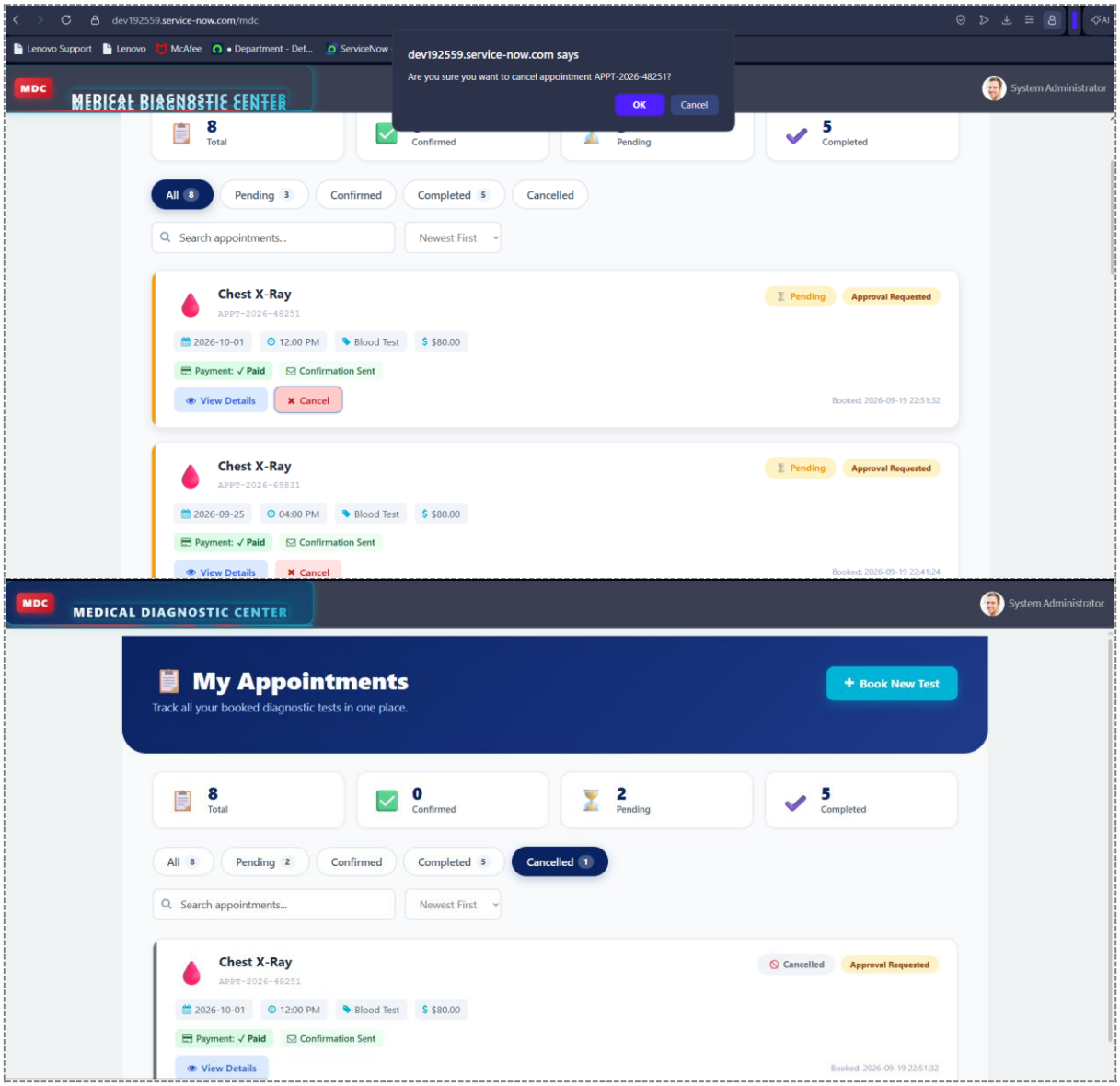


Figure 12: Business Rule – Automatic Report Generation

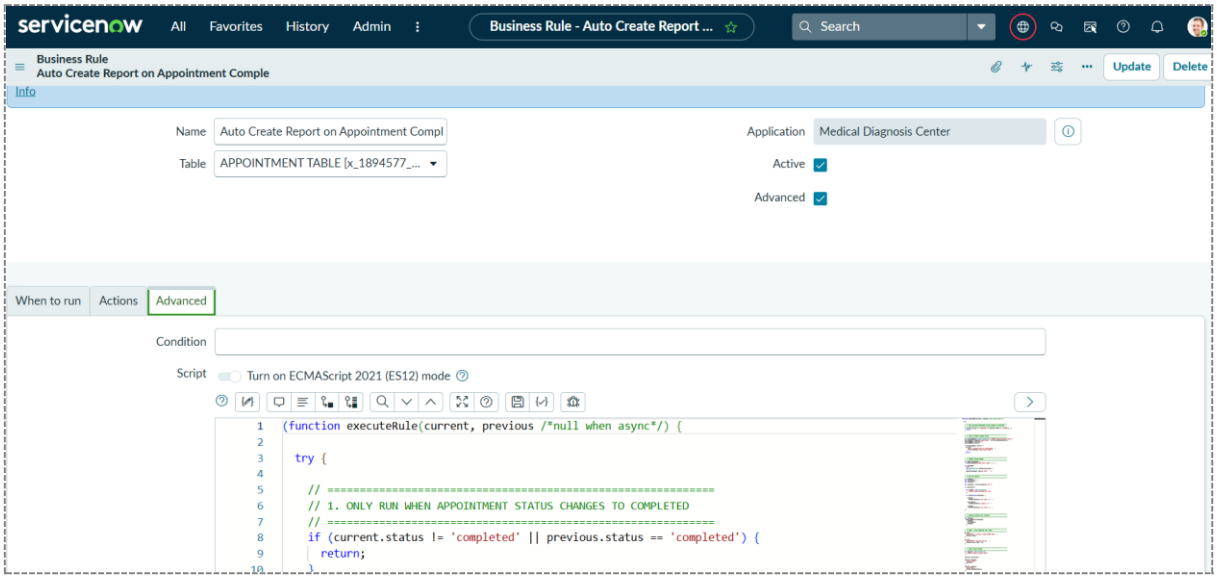


Figure 13: Notification / Email

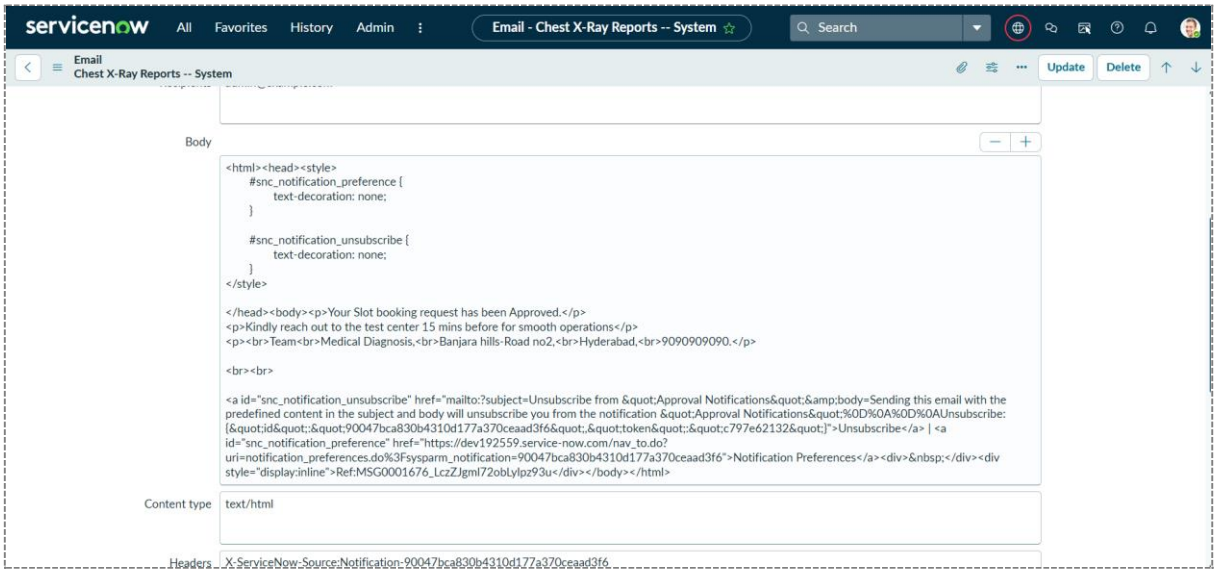
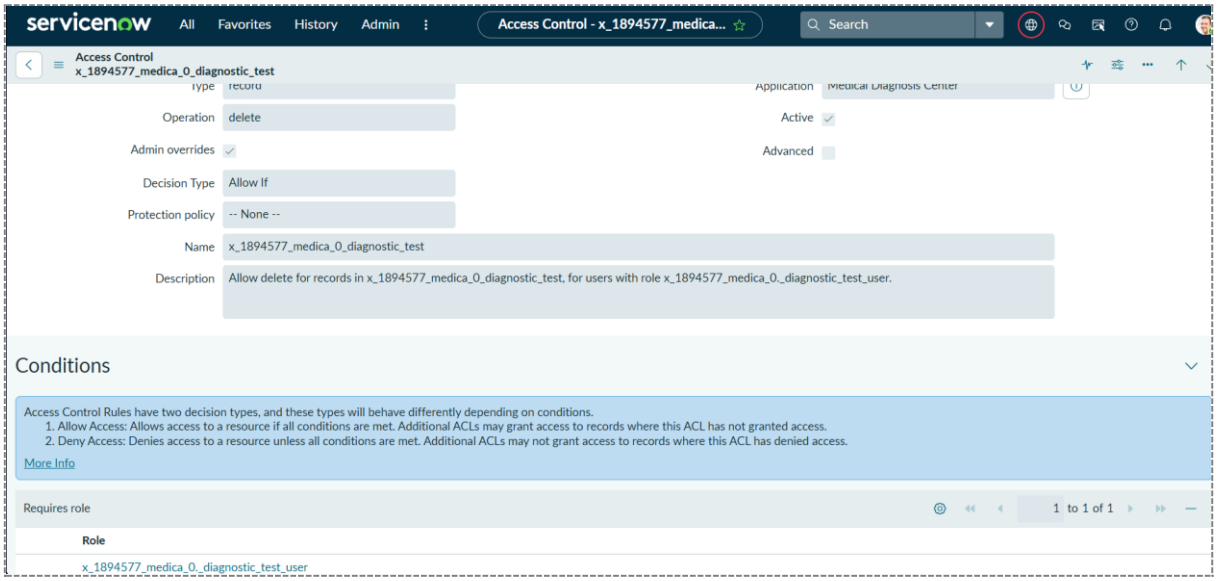


Figure 14: ACL / Role Configuration



12. Known Issues

The following limitations are documented based on the current implementation:

S.No.	Known Issue / Limitation	Status
1	The project does not integrate with an external payment gateway.	Current limitation
2	The application does not use an external healthcare/laboratory API.	Current limitation
3	Advanced analytics and predictive features are not implemented.	Future enhancement
4	Laboratory report download functionality depends on availability of a report file/URL.	Current limitation
5	The application is developed and tested within the ServiceNow PDI environment.	Current limitation

13. Future Enhancements

The following features can be added in future versions:

1. Real Payment Gateway Integration

Integrate a secure external payment gateway for real-time payment processing.

2. Advanced Analytics

Add dashboards for:

- Number of appointments
- Popular diagnostic tests
- Completed tests
- Report statistics
- Monthly booking trends

3. SMS Notifications

Add SMS notifications for appointment confirmation, reminders and report availability.

4. External Healthcare Integration

Integrate the application with external laboratory or healthcare systems through APIs.

5. Online Report Download

Provide downloadable PDF laboratory reports.

6. Doctor / Laboratory Staff Portal

Provide dedicated interfaces for doctors and laboratory technicians to update and verify test results.

7. Appointment Reminders

Automatically send reminders before the scheduled appointment.

8. Advanced Search

Allow patients to search reports and appointments using additional criteria.

Conclusion

The Medical Diagnostic Center – ServiceNow project provides a centralized digital solution for managing the diagnostic testing journey. The system integrates test browsing, appointment booking, payment status, appointment tracking, automated laboratory report generation, report viewing, notifications and access control within the ServiceNow platform.

The application was developed and tested using ServiceNow Service Portal, custom tables, Business Rules, Client Scripts, JavaScript, AngularJS and GlideRecord. The final UAT testing confirmed that the major application workflows functioned as expected.